# TUTORIAL PASO A PASO: CONFIGURACIÓN MAESTRA DE pfSense CE 2.9.0

### SIMULACIÓN DE INTERFAZ WebGUI & PARÁMETROS OFICIALES PARA KRONOS SENTINEL

**Autor:** Bruno Urrea Ortiz | Duoc UC Sede San Joaquín • **Portafolio de Título APT122** ($0 CLP)

## PASO 1: CONFIGURACIÓN DE TRONCAL 802.1Q Y CREACIÓN DE VLANs

Navegar a Interfaces > Assignments > VLANs y crear los 4 tags de subred sobre la interfaz física:

| ■ **pfSense WebGUI** // Interfaces / VLANs / Edit | |
|---|---|
| **Parent Interface** | vtnet1 (LAN Trunk Physical Adapter) [▼] |
| **VLAN Tag** | 10 (Corp) \| 20 (DMZ) \| 30 (VoIP) \| 99 (Mgmt) |
| **VLAN Priority** | 0 (Default 802.1p Best Effort) |
| **Description** | VLAN_10_CORP \| VLAN_20_DMZ \| VLAN_30_VOIP \| VLAN_99_MGMT |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 2: SERVIDOR DHCP Y RESERVAS ESTÁTICAS DE IP (DVWA Y ASTERISK)

Navegar a Services > DHCP Server > DMZ_SERVERS para configurar la subred y reserva fija:

| ■ **pfSense WebGUI** // Services / DHCP Server / DMZ_SERVERS (VLAN 20) | |
|---|---|
| **Enable DHCP Server** | [X] Enable DHCP server on DMZ_SERVERS interface |
| **Subnet & Range** | 192.168.20.0/24 \| Rango dinámico: 192.168.20.100 - .199 |
| **DNS & Gateway** | DNS: 192.168.20.1, 1.1.1.1 \| Gateway: 192.168.20.1 |
| **Static Mapping DVWA** | MAC: 02:42:c0:a8:14:32 ➔ IP Fija: 192.168.20.50 (dvwa-dmz) |
| **Static Mapping Asterisk** | VLAN 30 ➔ MAC: 02:42:c0:a8:1e:32 ➔ IP: 192.168.30.50 |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 3: SURICATA 7.X — MODO INLINE IPS NETMAP & EVE JSON

Navegar a Services > Suricata > Interfaces > Edit [WAN] y activar la inspección directa en hardware:

| ■ **pfSense WebGUI** // Services / Suricata / Interfaces / WAN Settings | |
|---|---|
| **Enable Inspection** | [X] Enable Suricata inspection on this interface |
| **Interface Selection** | WAN (vtnet0) [▼] |
| **IPS Mode Selection** | ( ) Legacy Mode (X) Inline Mode (netmap framework) |
| **Block on Alerts** | [X] Block offenders (Hardware ring-buffer packet drop) |
| **EVE JSON Logging** | [X] Enable EVE JSON Log \| Type: (X) FILE (/var/log/suricata/eve.json) |
| **EVE Event Types** | [X] Alerts (SQLi/RCE) [X] HTTP (URI, Host, Headers) |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 4: GESTIÓN DE FIRMAS ET OPEN Y POLÍTICA ATÓMICA dropsid.conf

Navegar a Services > Suricata > SID Mgmt para convertir alertas en drops inmediatos:

| ■ **pfSense WebGUI** // Services / Suricata / SID Mgmt / dropsid.conf | |
|---|---|
| **Automatic SID Mgmt** | [X] Enable Automatic SID State Management |
| **Drop SID List** | dropsid.conf (pcre:emerging-sql, pcre:emerging-exploit, pcre:community-web) |
| **State Order** | Drop > Enable > Disable |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 5: pfBlockerNG-devel — GEOIP MAXMIND Y LISTAS THREAT FEEDS

Navegar a Firewall > pfBlockerNG > IP > GeoIP para blindaje perimetral geográfico:

| ■ **pfSense WebGUI** // Firewall / pfBlockerNG / IP / GeoIP & Threat Feeds | |
|---|---|
| **MaxMind License Key** | 1024982 / ******************** (Cuenta Comunitaria Free $0) |
| **Top Spammers GeoIP** | CN, RU, IR, KP, VN, NG [▼] ➔ Action: (X) Deny Inbound |
| **IP Feeds (C2/Malware)** | FireHOL_L1 & Spamhaus_DROP ➔ Action: Deny Inbound (Cada 4 hrs) |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 6: HAPROXY 2.8+ — FRONTEND SSL Y STICK-TABLES ANTI-DOS

Navegar a Services > HAProxy > Frontend y configurar rate limiting en memoria RAM:

| ■ **pfSense WebGUI** // Services / HAProxy / Frontend / Edit: KRONOS_HTTPS_VIP | |
|---|---|
| **Frontend & Address** | KRONOS_HTTPS_FRONTEND \| WAN IP:443 (SSL Offloading) |
| **Stick Tables** | stick-table type ip size 100k expire 30s store http_req_rate(10s),http_err_rate(10s) |
| **Protección L7** | http-request deny deny_status 429 if { sc_http_req_rate(0) gt 100 } |
| **Default Backend** | DVWA_DMZ_POOL ➔ IP: 192.168.20.50:80 (Health Check: HTTP /) |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 7: TAILSCALE SUBNET ROUTER & REGLAS DE FIREWALL ZERO TRUST

Navegar a VPN > Tailscale y Firewall > Rules para aislar y conectar componentes:

| ■ **pfSense WebGUI** // VPN / Tailscale & Firewall / Rules / DMZ_SERVERS | |
|---|---|
| **Tailscale Subnet Router** | Auth Key: tskey-auth-*** \| Advertised Routes: 192.168.30.0/24 (VoIP) |
| **Regla DMZ 1 (DNS/NTP)** | PASS IPv4 UDP ➔ DMZ_SERVERS to * : 53, 123 |
| **Regla DMZ 2 (Aislamiento)** | DROP IPv4 * ➔ DMZ_SERVERS to LAN_CORP (V10) & MGMT (V99) |
| **Regla VoIP (Asterisk)** | PASS IPv4 UDP ➔ Softphone CISO to Asterisk: 5060, 10000-10100 |
| **Acciones WebGUI** | **[ Save ] [ Apply Changes ]** |

## PASO 8: VERIFICACIÓN EN CONSOLA FREEBSD (CLI DE DIAGNÓSTICO)

```
# 1. Verificar Suricata activo con hilos Netmap:
ps aux | grep suricata

# 2. Comprobar inserción atómica en la tabla en memoria snort2c:
pfctl -t snort2c -T show
pfctl -t snort2c -T test 198.51.100.100

# 3. Tail en vivo del log de eventos EVE JSON analizado por KRONOS Engine:
tail -f /var/log/suricata/eve.json
```